Episode-6

Tuesday, 20 October 2026 4:22 PM

SCISOFT ACADEMY

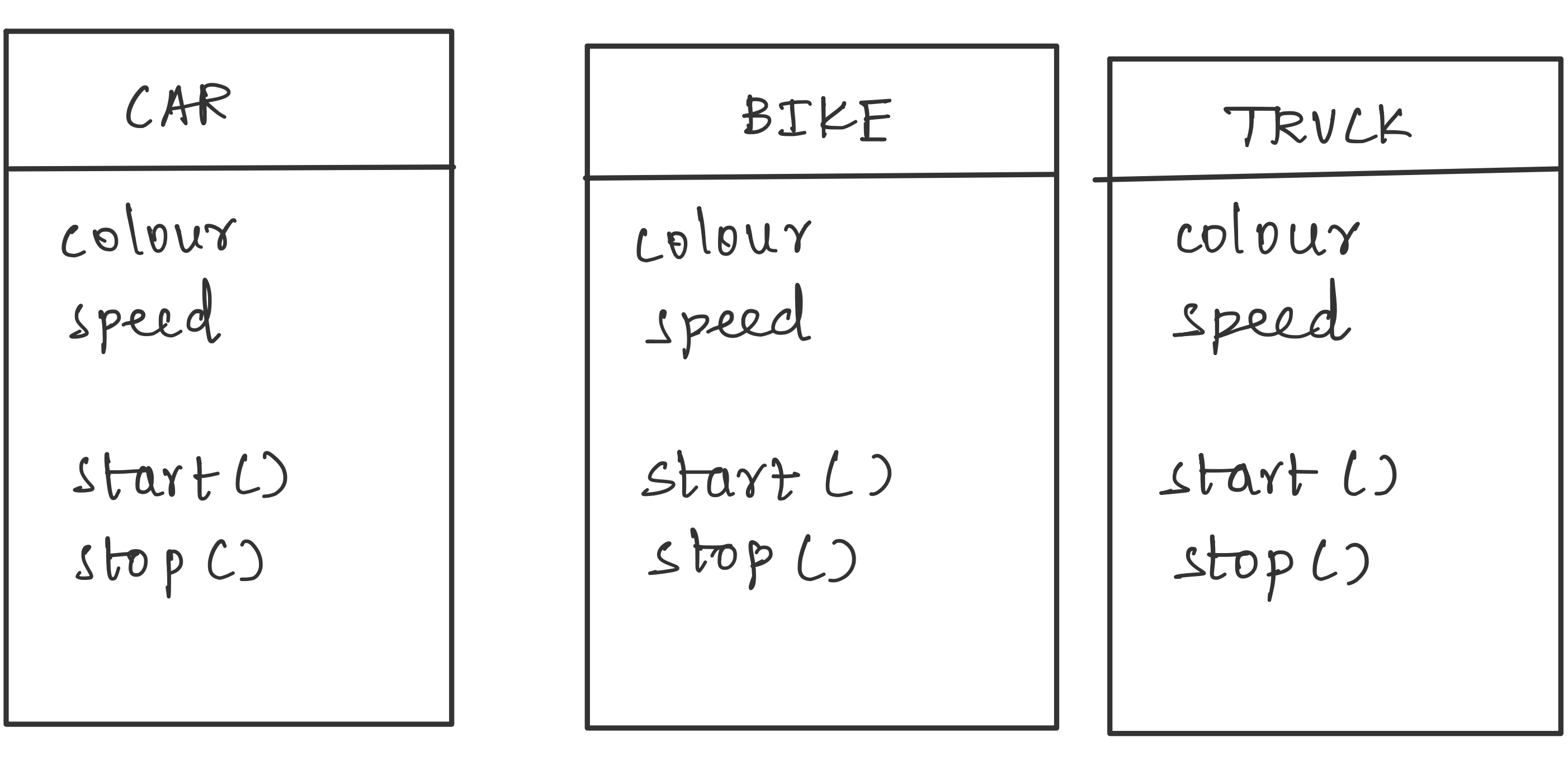
OBJECT ORIENTED PROGRAMMING

EPISODE -⑥

INHERITANCE

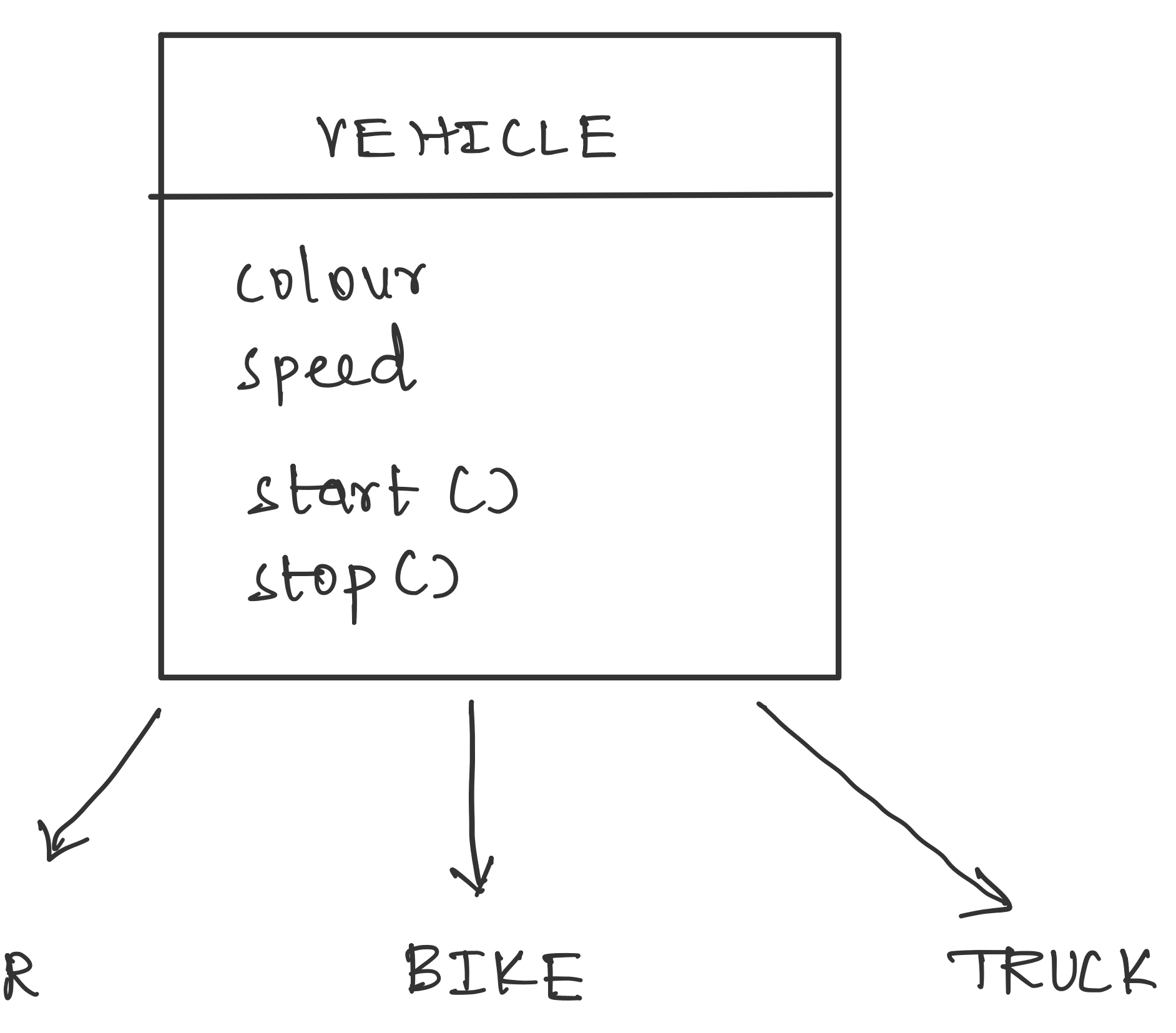
INHERITANCE → IS-A Relationship

PROBLEM: DUPLICATION

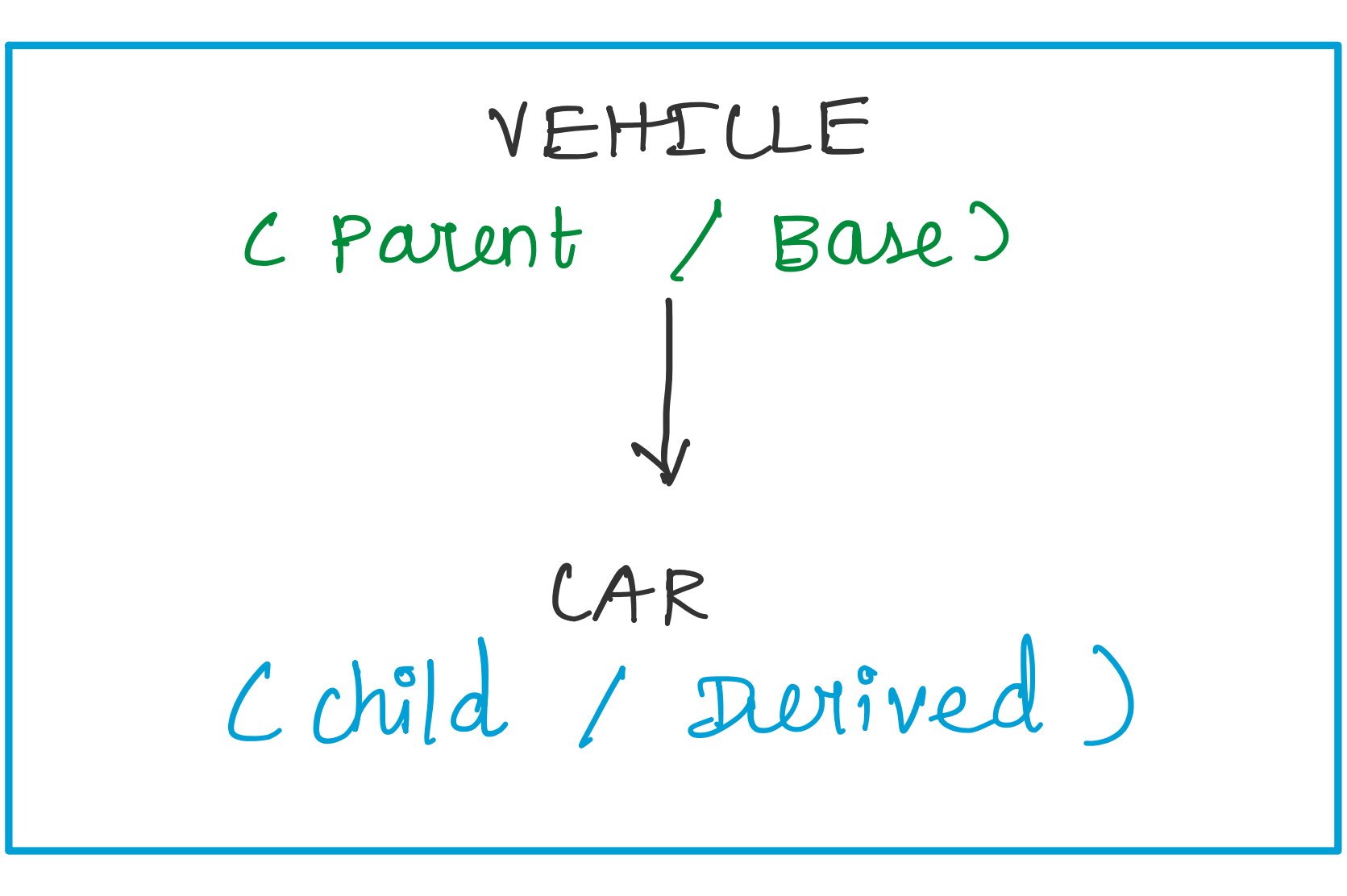


Duplication ???

Generalization:



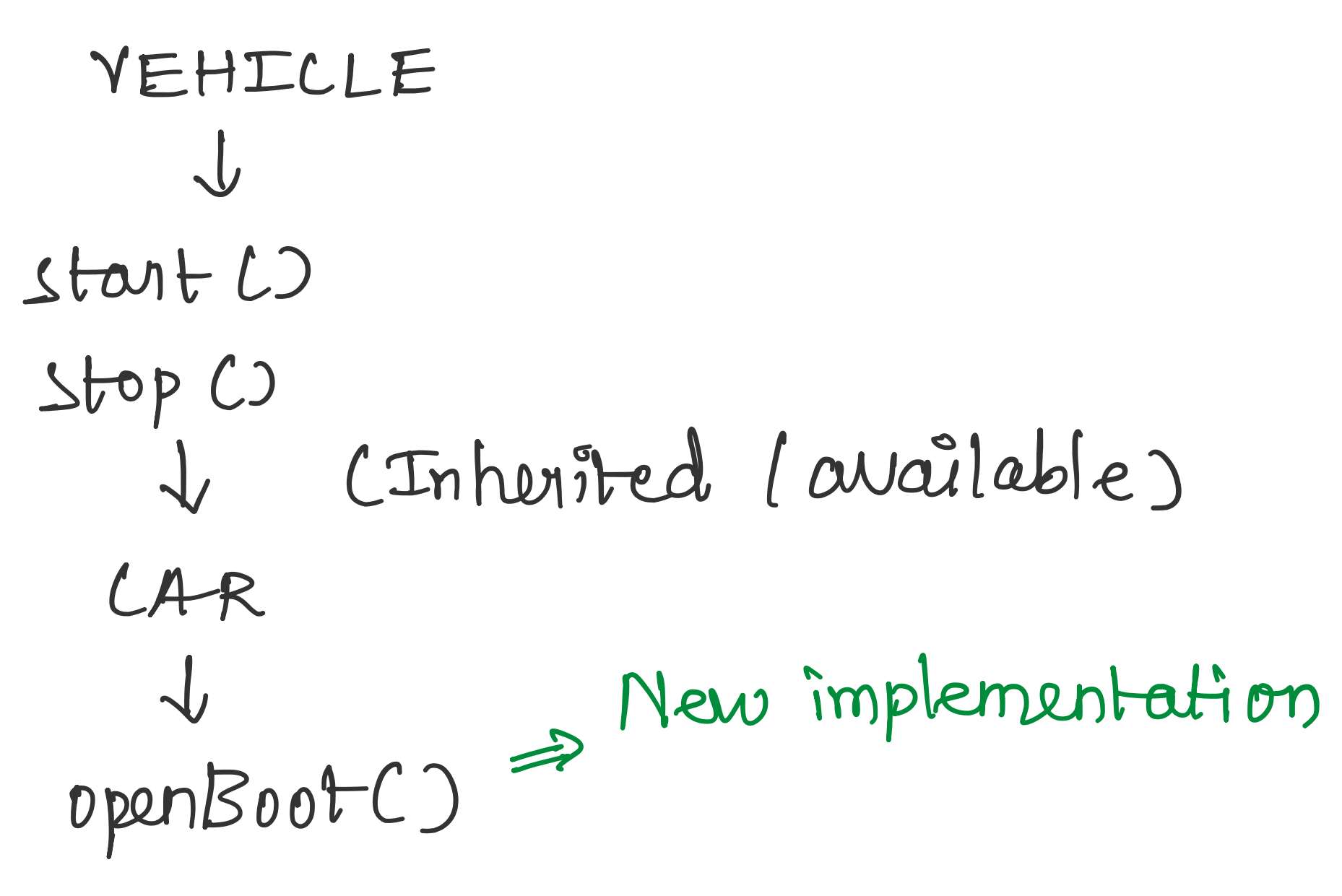
vehicle → General concept
Car → specialized vehicle
Bike → specialized vehicle
Truck → specialized Vehicle



- ✓ Car IS-A Vehicle
- ✓ Bike IS-A Vehicle
- ✓ Truck IS-A vehicle

IS-A

Inherited behaviour:

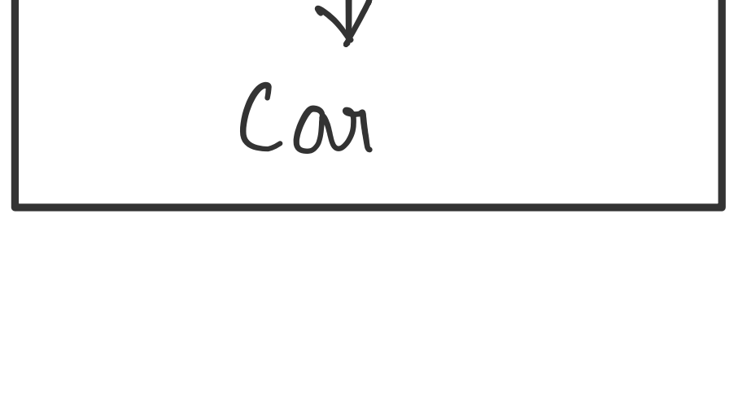


child can:

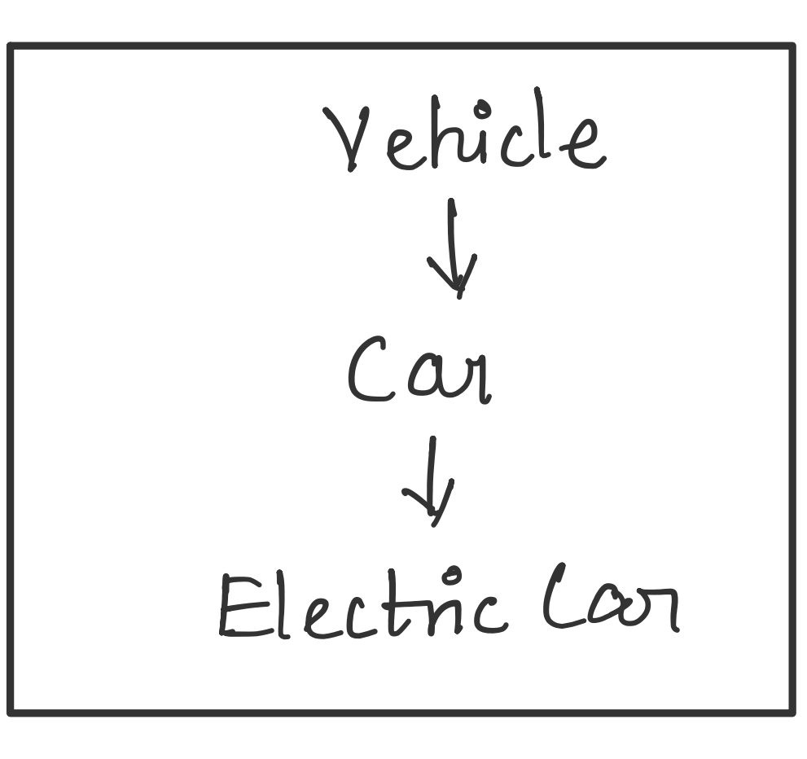
- ✓ Use inherited behaviour
- ✓ Add specialized behaviour
- ✓ Override suitable behaviour

Inheritance Types:

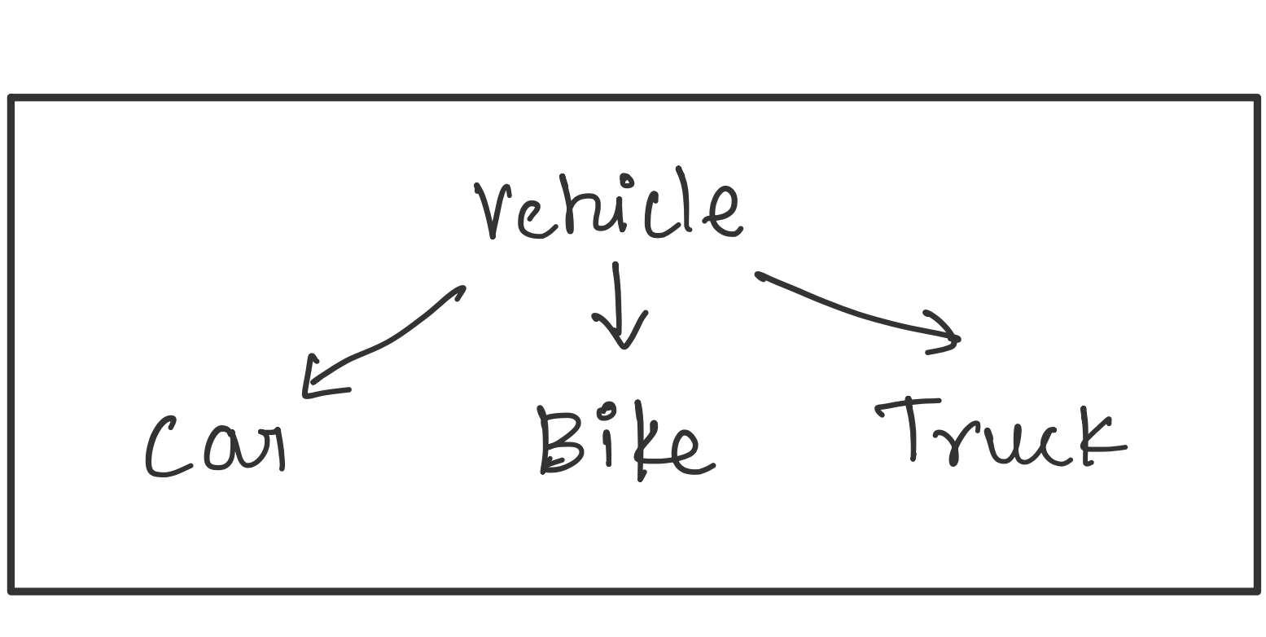
SINGLE



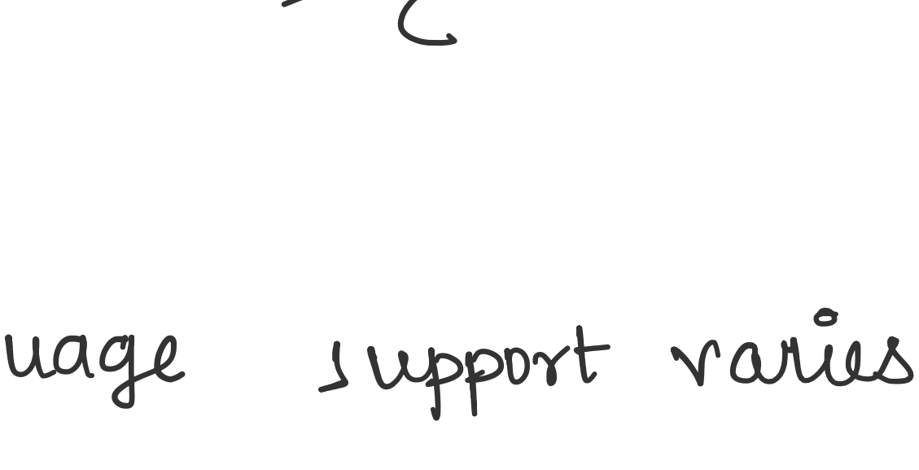
MULTI-LEVEL



HIERARCHICAL



MULTIPLE



Note: Language support varies.

Important design question:

IS-A vs **HAS-A**

Car IS-A Vehicle ✓
Car IS-A Engine ✗
Car HAS-A Engine ✓

What if the same operation behaves differently?
↓
POLYMORPHISM ...

NEXT VIDEO